

SECTION A

1 **(short)** A two-dimensional water jet with constant density ρ , and negligible friction is deflected in a right angle by a stationary circular obstacle as shown in Fig. 1. While in contact with the solid obstacle, the jet has circular streamlines between radii $r = R_1$ and $r = R_2$, with a velocity distribution $v(r) = \frac{C}{r}$.

- (a) Find the pressure difference Δp between the streamlines at $r = R_1$ and $r = R_2$ in terms of the constants ρ , C , R_1 , and R_2 . [6]
- (b) Find the components, parallel and perpendicular to the impinging jet, of the force by the jet on the obstacle, indicating their direction. [4]

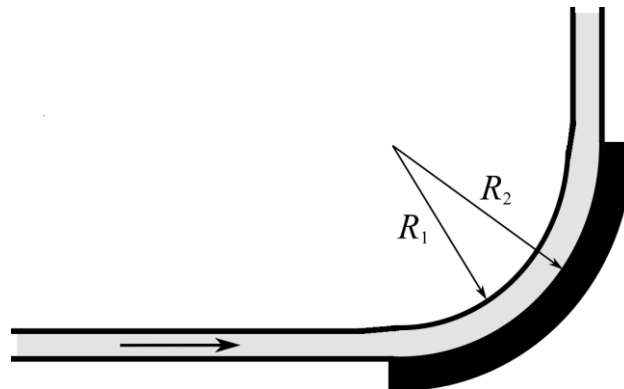


Fig. 1